Algebra Review: Unit 3 — Linear Equations & Slope

Answer Key

Station 1: Finding and Interpreting Slope

1. $m = \frac{4}{3}$ *points* $(-2, -1)$ *and* $(1, 3)$
2. $m = -1$
3. $m = 0$
4. undefined
5. $m = -\frac{2}{3}$ *points* $(-3, 2)$ *and* $(3, -2)$
6. $k = 1$ $\frac{10 - k}{3} = 3$
7. $m = 3$
8. $m = 2$ $y = 2x - 4$
9. $-\frac{12{,}000}{15} = -800$ ft/min
10. Yes. slope $= \frac{4}{2} = \frac{6}{3} = 2$ for both pairs

Station 2: Slope-Intercept Form

1. $m = -3,\ b = 7$
2. $m = \frac{2}{5},\ b = -4$
3. $y = -3x + 5;\ m = -3,\ b = 5$
4. $y = 4x - 2$
5. $y = -\frac{1}{2}x + 6$
6. $y = 3x - 1$ $5 = 3(2) + b$
7. $y = -2x + 2$ $m = \frac{-8}{4} = -2$
8. $y = \frac{3}{2}x - 1$
9. $y = -x + 2$ $b = 2$*;* $m = \frac{-1 - 2}{3 - 0} = -1$
10. $y = 2x + 3$; \$27

Station 3: Point-Slope and Standard Form

1. $y + 4 = 2(x - 3)$
2. $y - 1 = -\frac{1}{3}(x + 6)$
3. $y - 2 = 2(x - 1)$
4. $y = 3x - 1$
5. $y = -2x + 7$
6. $2x - y = 5$
7. $3x + 4y = 8$
8. x-int: $(5, 0)$; y-int: $(0, 3)$
9. x-int: $(6, 0)$; y-int: $(0, -4)$
10. $3x + y = 7$

Station 4: Parallel and Perpendicular Lines

1. 4
2. $-\frac{1}{4}$
3. $\frac{3}{2}$
4. $y = -2x + 3$
5. $y = -\frac{1}{2}x + 3$ $1 = -\frac{1}{2}(4) + b$
6. $y = 3x - 5$
7. Parallel *second line:* $y = 2x - 3$
8. Perpendicular *slopes* 3 *and* $-\frac{1}{3}$
9. Neither *slopes* -2 *and* $-\frac{1}{2}$
10. (a) $y = 7$ (b) $x = 5$